

6.2. Lineární operátor f na $V = \text{Span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\} \subseteq \mathbb{R}^4$

oplňuje

$$f((2, 1, 0, 1)^T) = (4, 1, 2, 5)^T \quad f((1, 0, 1, 2)^T) = (2, 2, -2, -2)^T$$

$$f^* \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = ? \quad B = \left(\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right)$$

$$1 \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + 2 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix} \quad \underline{\underline{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}}$$

$$2 \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + (-2) \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix} \quad \underline{\underline{\begin{pmatrix} 2 \\ -2 \end{pmatrix}}}$$

$$\Rightarrow [f]_B^B = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix} = A$$

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{pmatrix} = (\lambda+3)(\lambda-2)$$

$$\lambda_1 = -3 \quad \lambda_2 = 2$$

pl. sebou $\lambda_1 = -3$ $\begin{pmatrix} 1+3 & 2 \\ 2 & -2+3 \end{pmatrix} \sim \begin{pmatrix} 4 & 2 \\ 0 & 0 \end{pmatrix}$ $\text{Span} \left\{ \begin{pmatrix} -1 \\ 2 \end{pmatrix} \right\}$ $\mu_1 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$

$\lambda_2 = 2$ $\begin{pmatrix} 1-2 & 2 \\ 2 & -2-2 \end{pmatrix} \sim \begin{pmatrix} -1 & 2 \\ 0 & 0 \end{pmatrix}$ $\text{Span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$ $\mu_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$$+1 \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + 2 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix} \quad \text{line} = \left(\begin{pmatrix} 5 \\ 1 \\ 1 \\ 7 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix} \right)$$

$$2 \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + 1 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} \quad \text{ne} = \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = 1 \cdot \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} - 1 \cdot \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix}$$

$$f(v) = 2 \cdot \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} - (-3) \cdot \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix}$$